



Programming Fundamentals

Class Assignment # 1

Total Marks: 30 Instructor Name: Faiza Khadim

Deadline: 31-03-25 till 11:59 PM

Instructions:

- 1) You have to submit two files i.e.
 - CPP file of the program
 - CPP should be combined with files into zip folder.
- 2) You have to upload your assignments on CMS.
- 3) All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- 4) Plagiarism in the work is Professional dishonesty which leads to strict disciplinary action and zero in grading of Assignment.

<u>CLO NO</u>	<u>CLO STATEMENT</u>	<u>PLO</u>	<u>Taxonomy LEVEL</u>
2	Be able to apply fundamentals building blocks of a programming language to solve problems.	2	C3

Question 1:

[10-Points]

Write a program that reads the file sample.txt. The format of the file is full name, registration number, and obtained marks of Assignment, Quiz, Mid, Final from. Consider the following weightage

Graded Tasks	Total Marks	Weightage (%)
Assignments	20	15%
Quizzes	20	15%
Midterm	30	25%
Final Term	45	45%

Calculate the grade using above details, and consider UCP grading scheme. Now store the name, registration and grade in “result.txt”, on the basis of weightage. UCP grading scheme is given below.

Percentage (%)	Grade
85 - 100	A
80 - 84	A-
75 - 79	B+
70 - 74	B
65 - 69	B-
60 - 64	C+
55 - 59	C
50 - 54	C-
Below 50	F

Expected Output

sample.txt

Ali Ahmad, L1F17MSCS0011, 14, 14, 24, 43

result.txt

Ali Ahmad, L1F17MSCS0011, 79.7%, A-

Question 2: **[10-Points]**

Write a C++ program in which, read a **c-string word** from “**word_frequency.txt**”. Your task is to store the **word** with its **length** **separated by a coma** into another file “**word_lenght.txt**”. **Find the word which has max length on the console.**

Example:

word_frequency.txt

This course will also enable the students to develop practical applications

Word_length.txt

This ,4
Course ,6
will ,4
also ,4
enable ,6
the ,3
students ,8
to ,2
develop ,7
practical ,9
applications, 12

Max word: applications, 12

Question 3:**[10-Points]**

Write a C++ program in which, read **data** from “**Employee.txt**”. Your program should have many driven solution and user should press ‘Y’ to continue and press ‘N’ for terminate. You have to do the following tasks:

Employee.txt

Employee	Salary	Department
Ahsan Khan,	50000,	IT
Alia Shams,	60000,	Management
Najam ul Saqib,	30000,	Sales
Ahmer Javed,	50000,	HR

Expected Output:

*****Employee System*****

- Press ‘S’: Add more 5 records in the file with existing file format.
- Press ‘P’: Print all records on console.
- Press ‘E’: Find the employee which has max salary on the console.
- Press ‘D’: Department wise searching.
- Press ‘R’: Write all the record into another file except the one who is going to leave the department.
- Press ‘X’ : exit.

End of Assignment1 😊